

Technological University, Dublin

BSc. in Information Systems and Information Technology

Module Name: Programming and Algorithms 2

Module Code: CMPU2032

Jan-May Academic Term 2025

Problem-Based Assignment

Release Date: 16 April 2026

Submission Date: 3 May 2026 @ 23:59

Weight: The assignment will be marked out of 100. This assignment represents 30% of the overall marks for this module.

Instructions: This is an individual assignment.

SUBMISSION DETAILS: All the assignment deliverables must be submitted via Brightspace page before the deadline using the dedicated Brightspace submission pages. Late submissions will be penalised.

IMPORTANT: It is your responsibility to avoid plagiarism.

NOTE: YOU ARE NOT ALLOWED TO PUBLISH/SHARE YOUR SOLUTION WITH OTHERS. All work submitted should be your own.

Assignment Description

A hospital wants to digitize its patient handling system. You are hired to develop a Python-based system that efficiently manages patient records, emergency cases, and data processing using appropriate data structures and algorithms.

Problem Statement

Design and implement a **menu-driven Python program** that demonstrates the use of following requirements:

Functional Requirements

Task 1: Data Structures Implementation:

Linked List (Main Records)

- Store patient records using a **linked list**.
- Each node contains: Patient ID, Name, Age and Condition.
- Operations: Add Patient, View Patients, Delete Patient and Search Patient.

Queue (Emergency Patients)

- Use a **queue** for emergency cases (FIFO)
 - Operations: Add emergency patient, Process next emergency patient and Display queue.

Task 2: Sorting Algorithms:

Merge Sort: Sort patient records by **Patient ID**.

Shell Sort: Sort patient records by **Age**.

Menu Interface

===== Smart Hospital System =====

1. Add Patient
2. View Patients
3. Delete Patient
4. Search Patient
5. Add Emergency Patient
6. Process next Emergency Patient

7. View Emergency Queue
8. Sort by ID (Merge Sort)
9. Sort by Age (Shell Sort)
10. Exit

Enter your choice:

Task 3: Testing:

Students must design and include at least 5 test cases, covering:

- Normal cases (expected valid inputs)
- Edge cases (boundary or unusual inputs)

Task 4: Design Pattern:

Suggest at least two design pattern that can be implemented in your application and why the selected ones will be best option?

Assignment Deliverables

1. Submit a ZIP file containing source code.
2. A report with separate task-wise sections describing the implementation, including the following:
 - Explanation of the data structures used.
 - Description of sorting algorithms implemented.
 - Test cases.
 - Explanation of the design pattern selected.
 - Sample outputs of the program.
3. A short video (maximum 8 minutes) demonstrating and explaining your Python code, including execution of the menu-driven program. The URL of this video must be included in your report.

Assessment Criteria

The assignment will be assessed based on the marking rubric table shown below:

Component	Marks
Data Structures Implementation	20
Sorting Algorithms	20
Program Functionality	15
Testing (Task 3)	10
Design Pattern (Task 4)	5
Code Quality	5
Report + Video	25
Total	100

EXTRA CREDIT ASSIGNMENT

Note: This is an *extra credit assignment*; although it is graded out of 10 marks, the overall course total remains capped at 100, so the maximum achievable overall mark will still be 100.

Extra Credit (10 Marks) – SQLite CRUD

Task: Create a Python program using sqlite3 to perform basic **CRUD operations** for a employee management system.

Functional Requirements:

1. **Create table** Employees(id, name, department, salary) .. 2 marks
2. **Insert** at least 4 records .. 2 marks
3. **Display** all records on the terminal and DB Browser for SQLite..3 marks
4. **Update** at least two employees records .. 2 marks
5. **Delete** one employee .. 1 mark

Submission Requirement:

If attempting this extra credit assignment, submit your source code as a separate ZIP file along with a 2–3-minute demonstration video. A brief report (maximum 2 pages) is also required, along with the URL of the uploaded video.